

PROFESSIONAL SUMMARY

Data Scientist with an M.S. in Data Science (4.0 GPA, University at Buffalo) and hands-on production experience building end-to-end ML systems, anomaly detection pipelines, and NLP solutions. Skilled in statistical modeling, A/B testing, and feature engineering, with a strong foundation in Python, SQL, and experiment-driven decision-making. Experienced translating complex data into actionable business insights via dashboards and reporting. Seeking a Data Scientist role where I can drive measurable impact at scale.

TECHNICAL SKILLS

Statistical & ML Modeling: Scikit-learn, XGBoost, LightGBM, PyTorch, TensorFlow, A/B Testing, Hypothesis Testing

Data & Feature Engineering: Python, SQL (Postgres, MySQL), Pandas, NumPy, ETL Pipelines, PySpark, Hadoop

NLP & Generative AI: RAG, FAISS, SpaCy, Prompt Engineering, Explainable AI (XAI)

Experiment Tracking & MLOps: MLflow, Scalene, CodeCarbon, Docker, Model Profiling

Visualization & Reporting: Tableau, Power BI, Streamlit, Gradio, Matplotlib, Plotly

Cloud & Infrastructure: AWS S3/EC2, Linux/Bash, Git

EXPERIENCE

Troy & Banks

Data Scientist

Buffalo, NY, USA

Feb 2026 – Present

- Built and deployed an end-to-end ML system for utility bill extraction, tariff validation, and anomaly detection across 10,000+ PDFs and spreadsheets monthly, reducing manual audit effort by 40%.
- Designed scalable feature engineering and data ingestion pipelines; applied rule-based and ML-assisted logic (Isolation Forest, statistical thresholds) to detect demand and usage anomalies with > 90% precision.
- Developed Streamlit dashboards for real-time monitoring and client-ready reporting, cutting report turnaround time from days to hours.
- Collaborated cross-functionally with domain experts to translate business requirements into data-driven hypotheses and measurable KPIs.

Data Science and AI Intern

Aug 2025 – Dec 2025

- Scoped and prototyped the initial ML pipeline for bill anomaly detection.
- Conducted exploratory data analysis (EDA) and feature selection across structured and unstructured data sources to identify key statistical signals.

University at Buffalo

Research Assistant, Sustainable Machine Learning

Buffalo, NY, USA

Sept 2025 – Dec 2025

- Benchmarked large-scale models (NCF, Transformer-XL, GNMT, Mask R-CNN) across accuracy, runtime, memory, and energy efficiency metrics using Scalene and CodeCarbon.
- Supported reproducible experimentation and system-level analysis of deep learning model trade-offs, contributing to a research report on sustainable AI practices.

Sri Venkateswara College

Research Assistant

New Delhi, India

Jul 2021 – Nov 2021

- Applied regression modeling and sector-wise feature analysis to individual carbon footprint data, identifying key socioeconomic drivers of emissions.
- Presented findings at ICCIGE-2021 (International Conference on Climate Change — An Imminent Global Emergency).

PROJECTS

Credit Risk Modeling: Probability of Default — *Python, Scikit-learn, Gradient Boosting, Logistic Regression, Random Forest* Buffalo, NY

- Built a credit risk classification pipeline on the UCI German Credit dataset (1,000 applicants, 21 features) to predict probability of default (PD). Compared Logistic Regression, Random Forest and Gradient Boosting using 5-fold stratified cross-validation; Gradient Boosting achieved best test ROC-AUC of 0.8007 (CV: 0.783±0.034).
- Explored key risk factors (credit amount, loan duration, employment, housing), engineered features with encoding and scaling, and delivered top feature insights along with per-applicant PD scores for interpretability.

Gen AI in Education: Data Challenge — *Python, TensorFlow, NLP, Hugging Face*

Buffalo, NY

- Engineered an NLP pipeline for prompt generation and summarization to improve dataset diversity and labeling quality, achieving 30% increase in dataset diversity and 90.2% annotation precision. Received honorable mention from evaluation jury.

AI Radiology Report Summarizer — *RAG, RNN+Attention, GRU, TF-IDF, Streamlit*

Buffalo, NY

- Built an end-to-end retrieval-augmented NLP system: custom tokenization, XML parsing, GRU-based sequence modeling and TF-IDF retrieval on the OpenI clinical dataset. Deployed as a live Streamlit app for real-time inference.

Deep Learning for Network Traffic & Cybersecurity — *PyTorch, Sklearn, Plotly*

Buffalo, NY

- Implemented anomaly detection models (autoencoders, LSTMs, Transformer-based) on NSL-KDD and KDD'99 datasets; evaluated with ROC-AUC and Precision-Recall metrics to identify malicious traffic patterns.

EDUCATION

University at Buffalo (GPA: 4.0/4.0)

M.S. Data Science and Applications

Buffalo, NY

Dec 2025

- **Focus:** Machine Learning, Deep Learning, Statistical Modeling, Databases, AI Systems

Sri Venkateswara College, University of Delhi (GPA: 3.2/4.0)

B.S. (Hons) Mathematics

New Delhi, India

May 2023

- **Focus:** Probability, Statistics, Linear Algebra, Optimization, Real Analysis

CERTIFICATIONS

Microsoft Power BI Desktop for Business Intelligence (2023) — [Link](#)

SQL — MySQL for Data Analytics and Business Intelligence — [Link](#)